



Tanisha Sonkar

Bachelor of Technology
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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech., EE	Indian Institute of Technology, Jodhpur	7.28	2023-2025
Senior Secondary	Ann Mary School, Dehradun	94	2023
Secondary	Ann Mary School, Dehradun	98	2021

PROJECTS

Multilingual Music Recommender Web App

Mar 2025 – Apr 2025

Tools: React, Flask, Scikit-learn

[\[GitHub\]](#) [\[Project Page\]](#)

- Built a multilingual recommender system for 60K+ songs using **Kernel PCA** and **K-Means** clustering to model genre and mood similarity.
- Used **cosine similarity** within language-specific clusters to deliver context-aware recommendations.
- Developed an interactive frontend in React and full-stack application with Flask backend and real-time visualizations.

AI Overstock Liquidation Dashboard

Jun 2025 – Present

Tools: React, Flask, MongoDB Atlas

[\[GitHub\]](#)

- Built an AI-driven dashboard to forecast and clear overstock across **42K+ SKUs** using Walmart's M5 dataset.
- Achieved RMSLE of 0.40 on 1.8M+ sales records using LightGBM/CatBoost with advanced time-series features (lags, zero streaks, price shifts).
- Applied **KMeans clustering** to group SKUs into smart clearance bins for targeted discount strategies.
- Integrated **MongoDB Atlas** for real-time store-level inputs, predictions, and campaign data storage.
- Designed an interactive **React + Flask** dashboard with bulk predictions and auto-clearance generator across 10+ stores.

FastICA Audio Source Separation System

Mar 2025 – Apr 2025

Tools: React, FastAPI

[\[GitHub\]](#)

- Developed a full-stack web application for Blind Source Separation using the FastICA algorithm to extract independent audio sources from mixed .wav inputs.
- Built a FastAPI backend to process uploaded signals, apply ICA via scikit-learn, and generate separated .wav files and waveform visualizations using NumPy + matplotlib.
- Implemented responsive React + Tailwind UI for uploading audio, viewing signal plots (mixed vs separated), and downloading outputs.
- Explored a custom FastICA implementation in Jupyter to understand convergence behavior and contrast function effects.
- Undertaken as a Design Credit project under the guidance of Prof. Rajendra Nagar, Dept. of Electrical Engineering.

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, HTML/CSS
Frontend: React, Tailwind CSS, Next.js, Three.js
Backend: Flask, Node.js, Express.js
Libraries & ML: scikit-learn, NumPy, Pandas, Matplotlib
Database: MongoDB
Tools: Git, VS Code

KEY COURSES TAKEN

Data Structures & Algorithms, Machine Learning, Signals & Systems, Introduction to Computer Science, Probability & Statistics